INCH-POUND

MIL-DTL-55041/11A
w/AMENDMENT1
21 March 2022
SUPERSEDING
MIL-DTL-55041/11A
2 October 2006

DETAIL SPECIFICATION SHEET

SWITCHES, WAVEGUIDE, 1P2T, LATCHING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switches described herein shall consist of this specification sheet and MIL DTL 55041.

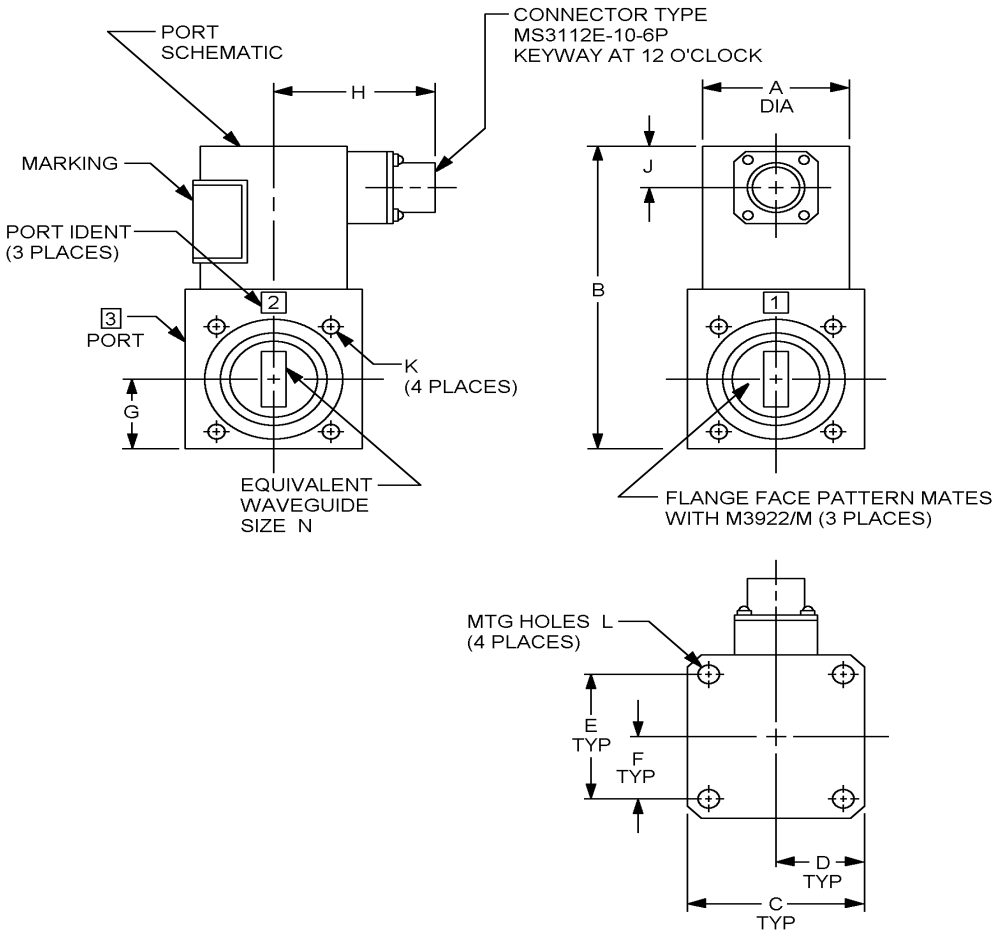


FIGURE 1. Switch configuration and schematic, part numbers M55041/11-001 through -005.

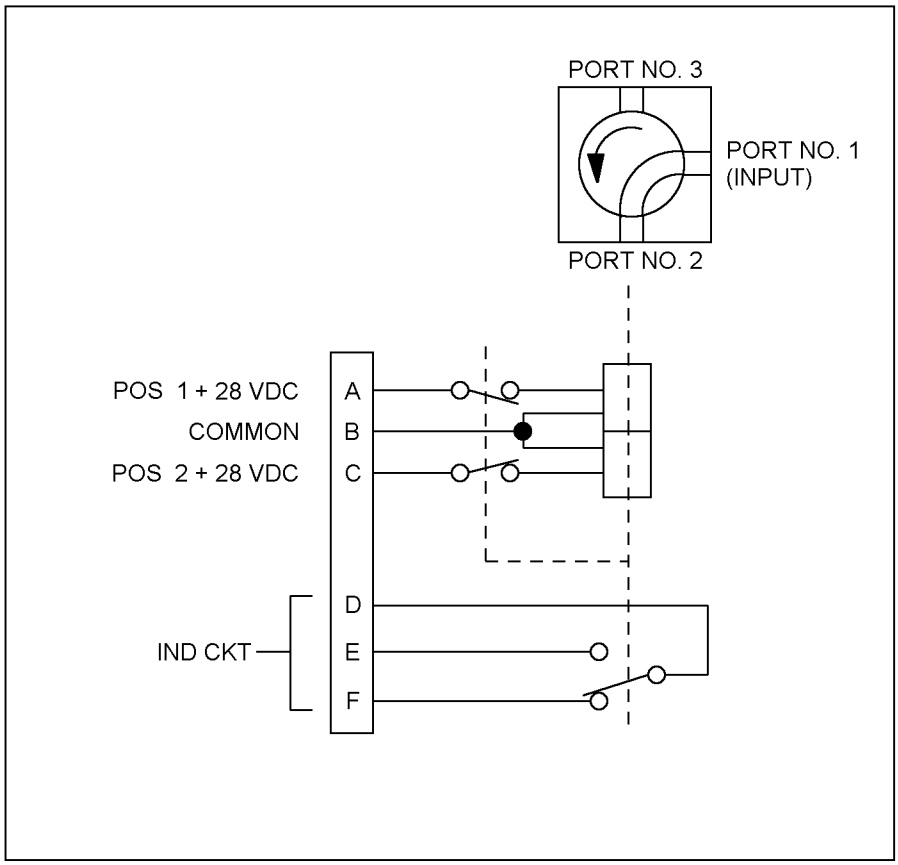
AMSC N/A

FSC 5985

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SCHEMATIC



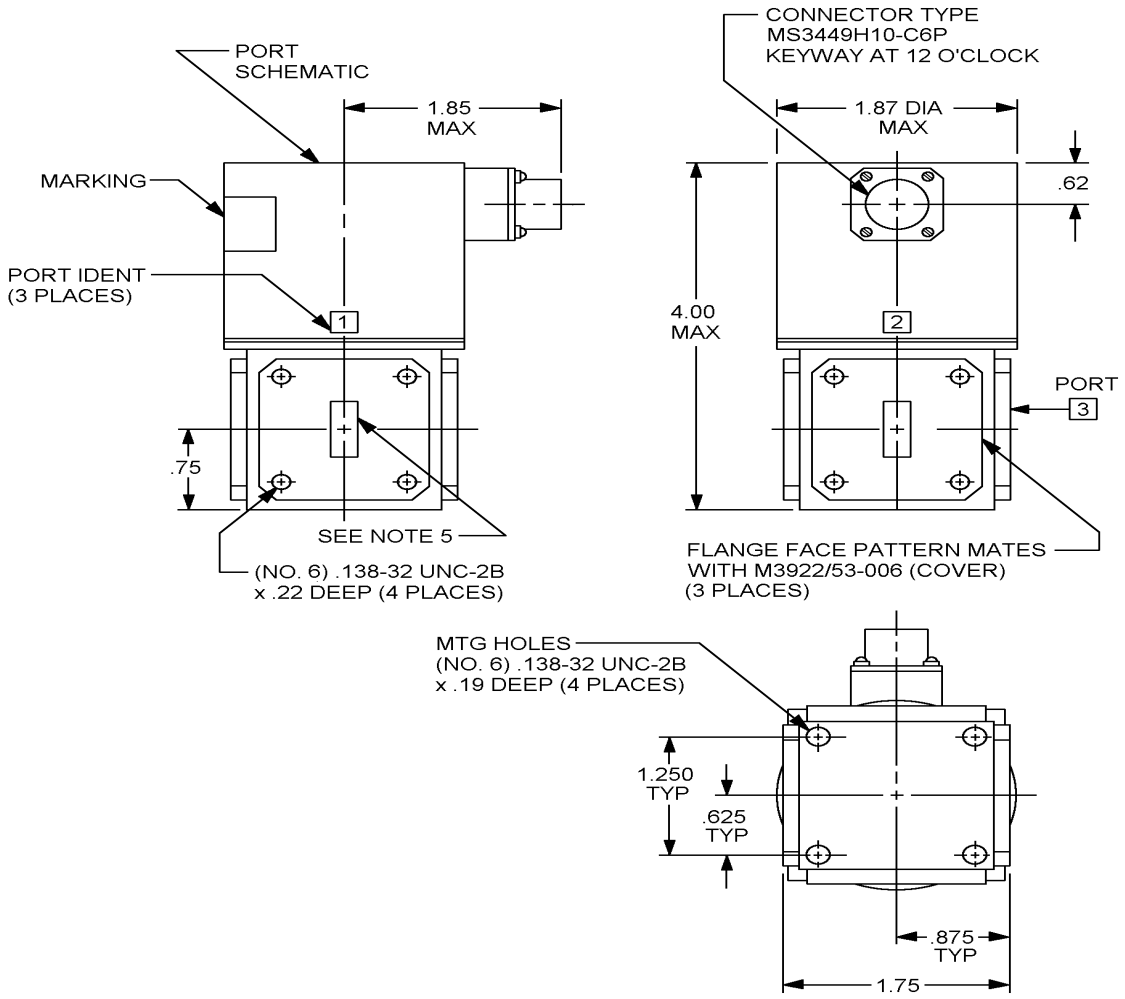
SHOWN IN LATCHING POSITION (POSITION 1)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ± .010 inch (0.25 mm) for three place decimals and ± .03 inch (0.8 mm) for two place decimals.
5. DC polarity must be observed.

FIGURE 1. Switch configuration and schematic, part numbers M55041/11-001 through -005 - Continued.

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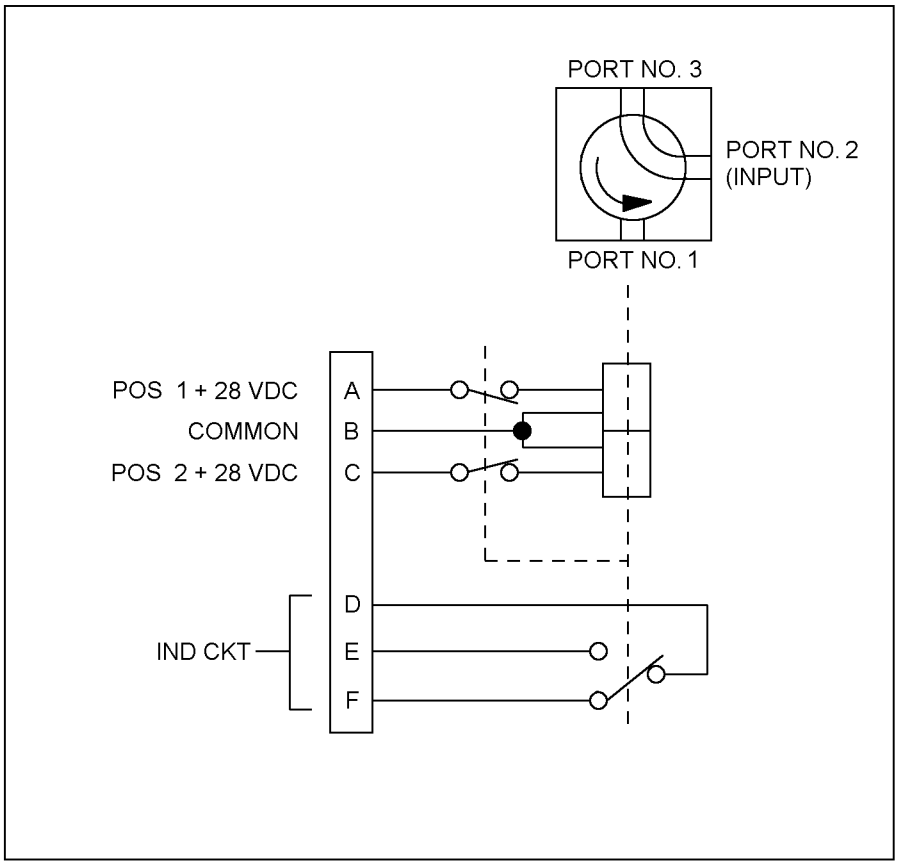


Inches	mm
.138	3.50
.19	4.8
.22	5.6
.62	15.7
.625	15.8
.75	19.05
.875	22.2
1.250	31.7
1.75	44.5
1.85	46.9
1.87	47.5
4.00	101.6

FIGURE 2. Switch configuration and schematic, part number M55041/11-006.

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SCHEMATIC



SHOWN IN LATCHING POSITION (POSITION 1)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (in millimeters) are given for general information only.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ± .010 inch (0.25 mm) for three place decimals and ± .03 inch (0.8 mm) for two place decimals.
5. DC polarity must be observed.

FIGURE 2. Switch configuration and schematic, part numbers M55041/11-006 - Continued.

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TABLE I. Dash numbers and dimensions.

Dash number	A max	B ±.03	C ±.03	D ±.010	E ±.010	F ±.010	G ±.010	H max	J max	K	L	M 1/	N
001	3.25	9.00	5.88	2.940	4.750	2.375	2.750	2.57	0.65	.250-20 x .40 deep	.250-20 x .50 deep	M3922/ 56-002	M85/1-041
002	3.25	7.50	4.25	2.125	3.500	1.750	2.000	2.57	0.65	.190-32 x .31 deep	.250-20 x .40 deep	M3922/ 57-001	M85/1-053
003	3.25	7.00	4.25	2.125	3.500	1.750	1.750	2.57	0.65	.250-20 x .40 deep	.190-32 x .30 deep	M3922/ 55-002	M85/1-065
004	2.25	5.10	2.38	1.190	2.000	1.000	1.245	2.07	0.65	.164-32 x .31 deep	.164-32 x .28 deep	M3922/ 53-003	M85/1-078
005	1.85	4.00	1.88	0.94	1.437	0.719	0.877	1.87	0.65	.138-32 x .22 deep	.164-32 x .25 deep	M3922/ 53-006	M85/1-091

1/ PINs are from MIL-DTL-3922/53, /55, /56, and /57.

TABLE II. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Position indicator circuit	Interlock circuit	Nominal operating range	Pull-in voltage (less than)	Operating current (max) 3/
			GHz	amperes 3/		V dc	V dc	amperes
001	1	1/ 2/	2.60 - 3.95	0.1	N/A	28	20	1.5
002	1	1/ 2/	3.95 - 5.85	0.1	N/A	28	20	1.0
003	1	1/ 2/	5.85 - 8.20	0.1	N/A	28	20	1.0
004	1	1/ 2/	7.05 - 10.0	0.1	N/A	28	20	1.0
005	1	1/ 2/	12.4 - 18.00	0.1	N/A	28	20	1.0
006	2	1/ 2/	12.4 - 18.00	0.1	N/A	28	20	2.0

TABLE II. Electrical performance characteristics - Continued.

Dash number	Transient interference provisions	Isolation (min)	VSWR (max)	Insertion loss (max)	Switching time	Waveguide pressuri-zation	RF power handling capability (peak)	Life cycles (X1,000)	Weight (max)
		dB		dB	ms	lbf/in^2	kW duty cycle		
001	yes	60	1.10:1	0.08	500	20	2200 33×10^{-3}	200	17 lbs
002	yes	60	1.10:1	0.08	500	20	1400 31×10^{-3}	200	9 lbs
003	yes	60	1.10:1	0.1	150	20	560 26×10^{-3}	200	8.3 lbs
004	yes	60	1.10:1	0.15	100	20	275 27×10^{-3}	200	2.3 lbs
005	yes	60	1.10:1	0.15	100	20	120 24×10^{-3}	200	1.3 lbs
006	yes	60	1.10:1	0.1	125	45	43 1×10^{-3}	500	1.5 oz

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.
2/ Rotor and housing of aluminum alloy.
3/ At 28 V dc and 20°C to 25°C.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1, figure 2, and table I.

Power connectors: See figure 1 and figure 2.

Operation: Break before make.

Operating temperature range: -54°C to +95°C.

Electrical performance characteristics: See table II.

Part or identifying number (PIN): M55041/11- (and dash number from tables I and II).

Substitutability data: The switches covered by this specification sheet are substitutable for the user's or manufacturer's PIN specified in table III. This information in no way implies that the manufacturer's PIN or NSN is suitable as a substitute for the military PIN.

TABLE III. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985-
001	00471	33D08400	01-073-1529
002	00471	33D06400	01-235-7794
003	00471	33D04400	01-233-2561
004	81755	654VE4826-3	01-099-6113
004	00471	33D03400	01-099-6113
005	81755	C4715-1	00-009-3159
005	00471	33D00400	00-009-3159

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Referenced documents. In addition to MIL-DTL-55041, this specification sheet references the following documents:
MIL-DTL-3922/53
MIL-DTL-3922/55
MIL-DTL-3922/56
MIL-DTL-3922/57
MS3112
MS3449

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - SM, MI
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
DLA - CC

(Project 5985-2022-010)

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